3D Representation

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Working slogan: Shared Anatomy, Without Shared Topology.

1 Anatomy-Grounded Volumetric Footwear Representation

Our goal is to represent footwear instances that may differ radically in geometry, material support, and topology while retaining a shared functional organization induced by human anatomy. A sneaker, sandal, clog, slipper, high heel, and boot cannot in general be registered to a common mesh: surface regions may appear or disappear, holes may change topology, and structures such as straps or heel enclosures may exist only for selected footwear types. Nevertheless, all footwear is organized around the same underlying human anatomy. We therefore separate *the anatomical coordinate system* from *the footwear material geometry*. The former provides a shared functional domain; the latter is allowed to vary freely in topology.

Our representation has two components:

$$\boxed{\text{Anatomical Volumetric Coordinates} + \text{Structured Footwear Material Field.}}$$

1.1 Shared Anatomical Volumetric Coordinates

Let

$$F_i = F(\boldsymbol{\beta}_i, \boldsymbol{\theta}_i) \subset \mathbb{R}^3$$

denote the parametric foot associated with footwear instance i, instantiated from SUPR-Foot. Different instances may use different foot shapes and canonical poses, but all are mapped to the same canonical anatomical domain $\mathcal{A}$ associated with a reference foot $F_0 = F(\boldsymbol{\beta}_0, \boldsymbol{\theta}_0)$.

We define a volumetric parameterization

$$\chi_i : \mathcal{A} \to \mathcal{N}(F_i),$$

where $\mathcal{N}(F_i)$ is the region surrounding the foot in which footwear material may occur. Its inverse

$$\Phi_i = \chi_i^{-1}, \qquad \mathbf{q} = \Phi_i(\mathbf{x}) = (u, v, r),$$

maps a Euclidean point $\mathbf{x}$ into shared anatomical coordinates. Here (u, v) denotes a functional position relative to the canonical foot and r parameterizes position through the surrounding anatomical volume. We use $r = 0$ at the canonical foot boundary and take positive r to extend outward into the footwear volume; deeply interior points of the foot are outside $\mathcal{N}(F_i)$ and are not part of the footwear domain.

Thus, the same coordinate (u, v, r) has a shared interpretation across instances—for example, a location below the plantar heel, lateral to the forefoot, or above the dorsal toe region—even when the footwear surfaces occupying that location differ or do not exist.

A nearest-surface parameterization is an important baseline but is not sufficient in general. If

$$\mathbf{p}^* = \operatorname*{argmin}_{\mathbf{p}\in F_i} \|\mathbf{x} - \mathbf{p}\|_2$$

and $\psi_i : F_i \to \mathcal{U}$ denotes a surface chart, a naive mapping would take the form

$$\Phi_i^{\text{near}}(\mathbf{x}) = (\psi_i(\mathbf{p}^*),\ \operatorname{sgn}(\mathbf{x})\,\|\mathbf{x} - \mathbf{p}^*\|_2)\,.$$

Such a mapping becomes unstable near concavities, between articulated toes, and far from the foot, where large footwear structures may collapse onto a small surface region. Our desired volumetric map should instead be approximately injective over the footwear-support region and have bounded distortion. In particular, for $\mathbf{x}_1, \mathbf{x}_2$ in the valid domain, we seek constants $0 < c < C < \infty$ such that

$$c\|\mathbf{x}_1 - \mathbf{x}_2\|_2 \le \|\Phi_i(\mathbf{x}_1) - \Phi_i(\mathbf{x}_2)\|_2 \le C\|\mathbf{x}_1 - \mathbf{x}_2\|_2.$$

This criterion directly distinguishes a useful anatomical volume from a nearest-point heuristic that folds or collapses space.

The canonicalization convention is part of the coordinate definition. In particular, heel elevation is mapped to a canonical plantar-flexed foot pose before constructing Φ_i, preventing heel height from being absorbed arbitrarily into either foot pose or shoe geometry. For boots, the anatomical domain is extended continuously from the foot through the ankle and lower-leg portion of SUPR. The specific construction of χ_i is evaluated in Sec. **??**; candidate implementations include harmonic volumetric coordinates and anatomical tetrahedral cages.

1.2 Structured Material Field: Where Anatomy Becomes a Prior

A coordinate transformation alone does not constitute an anatomical prior: with an unconstrained decoder, a sufficiently expressive network could treat Φ_i merely as a reparameterization. We therefore place anatomical structure explicitly inside the footwear representation.

For each instance, we first define a coarse *anatomical base envelope* using three functions over the canonical foot domain:

$$g_i(u, v), \qquad \delta_i^{\text{in}}(u, v), \qquad \delta_i^{\text{out}}(u, v).$$

The signed coverage field g_i determines where a primary footwear shell exists, with

$$g_i(u, v) < 0 \quad \Longleftrightarrow \quad \text{the anatomical region is covered,}$$

while δ_i^{in} and δ_i^{out} describe the cavity-facing and exterior extents along the anatomical volume, with

$$0 \le \delta_i^{\text{in}}(u, v) < \delta_i^{\text{out}}(u, v).$$

A corresponding base implicit field can be written as

$$B_i(u, v, r) = \max\left\{g_i(u, v),\ \delta_i^{\text{in}}(u, v) - r,\ r - \delta_i^{\text{out}}(u, v)\right\}.$$

Hence $B_i < 0$ describes the primary material volume surrounding the foot.

The anatomical prior enters explicitly through the inner envelope. We write

$$\delta_i^{\text{in}}(u,v) = \mu_{\text{clr}}(u,v;F_i) + \Delta_i^{\text{in}}(u,v),$$

where μ_{clr} is an anatomy-conditioned clearance prior and Δ_i^{in} is an instance-specific residual. Regularizing Δ_i^{in} toward zero makes a foot-shaped cavity the default explanation rather than something the decoder must rediscover independently for every shoe.

The base envelope captures the strongest shared structure, but it cannot alone represent arbitrary footwear. We therefore introduce a residual material field

$$R(\mathbf{q}, \mathbf{z}_i)$$

that can locally add or remove material. The complete implicit field is

$$f_i(\mathbf{x}) = B_i(\Phi_i(\mathbf{x})) + R(\Phi_i(\mathbf{x}), \mathbf{z}_i)\,,$$

with a sparsity or magnitude penalty on R encouraging the anatomical base envelope to explain geometry whenever possible while leaving the residual free to model straps, perforations, heel blocks, decorative elements, and other topology-changing structures.

The footwear material volume is

$$\Omega_i = \left\{\mathbf{x} \in \mathbb{R}^3 \mid f_i(\mathbf{x}) < 0\right\},$$

and the complete footwear surface is its boundary,

$$\mathcal{S}_i = \partial\Omega_i = \{\mathbf{x} \mid f_i(\mathbf{x}) = 0\}\,.$$

This material-volume interpretation is central. The exterior surface, the cavity-facing interior surface, outsole thickness, straps, and disconnected components are all different portions of the boundary of one occupied material volume. Holes in a clog or open regions of a sandal are simply locations where $f_i > 0$. No shared mesh connectivity is assumed.

Although the decoder operates in anatomical coordinates, we regularize f_i in Euclidean space, e.g.,

$$\mathcal{L}_{\text{eik}} = \mathbb{E}_{\mathbf{x}} \left(\|\nabla_{\mathbf{x}} f_i(\mathbf{x})\|_2 - 1\right)^2,$$

so that the learned field approximates a signed distance field in physical space rather than merely in the distorted anatomical coordinate domain.

This signed formulation is deliberate. UDF representations are attractive for idealized zero-thickness open surfaces, such as cloth sheets, because a global inside/outside distinction is undefined. Physical footwear components instead have finite material thickness and therefore define a valid occupied volume Ω_i with a meaningful sign. A signed material field consequently preserves cavity structure and supports standard level-set extraction. If extremely thin structures later violate this assumption, they can be handled by an auxiliary thin-structure or metric–phase formulation without changing the core anatomical representation.

1.3 Set-Valued Functional Correspondence

The shared anatomical domain induces correspondence without requiring shared mesh topology. Importantly, correspondence is not necessarily one-to-one. For a fixed functional coordinate (u, v), consider the anatomical fiber

$$\gamma_i^{u,v}(r) = \chi_i(u, v, r).$$

Its intersections with the footwear surface are

$$\mathcal{R}_i(u, v) = \{r \mid f_i(\chi_i(u, v, r)) = 0\}.$$

The corresponding set of physical surface locations is

$$\mathcal{C}_i(u, v) = \{\chi_i(u, v, r) \mid r \in \mathcal{R}_i(u, v)\}.$$

This set may be empty, contain a single element, or contain multiple elements. For example, a sandal may have no material over much of the dorsal foot, yielding

$$\mathcal{C}_i(u, v) = \varnothing,$$

whereas a layered region containing both an upper and a strap may produce multiple intersections. This set-valued formulation therefore treats absence and multiplicity as properties of footwear structure rather than correspondence failures. Two shoes correspond functionally by comparing $\mathcal{C}_i(u, v)$ and $\mathcal{C}_j(u, v)$ at the same anatomical coordinate.

1.4 Appearance and Completion

Geometry and appearance are separated. Given an appearance latent $\mathbf{z}_i^{\mathrm{app}}$, an appearance field

$$A_i(\mathbf{x}) = H(\Phi_i(\mathbf{x}), \mathbf{z}_i^{\mathrm{app}})$$

predicts surface appearance or material parameters at locations on $\mathcal{S}_i$.

Under ring-view capture, the latent representation is not known a priori and must be inferred from partial observations. Let $\mathcal{O}_i$ denote the available ring observations and let $\mathbf{z}_i$ collect the latent footwear variables. Completion is therefore posterior inference,

$$p\Big(\mathcal{S}_i^{\mathrm{hid}} \mid \mathcal{O}_i, F_i\Big) = \int p\Big(\mathcal{S}_i^{\mathrm{hid}} \mid \mathbf{z}_i, F_i\Big)\, p(\mathbf{z}_i \mid \mathcal{O}_i, F_i)\, d\mathbf{z}_i.$$

The uncertainty of unseen geometry therefore resides in the posterior over footwear latents rather than being hidden by conditioning on a known identity code. The anatomical base envelope constrains quantities such as cavity location, plantar support, and clearance, whereas the learned footwear latent captures design-specific structure such as outsole geometry, heel form, and other style-dependent regions.

In summary, the representation separates three roles:

$$\boxed{\underbrace{\Phi_i}_{\text{shared functional coordinates}} \quad + \quad \underbrace{B_i}_{\text{anatomical structural prior}} \quad + \quad \underbrace{R(\cdot, \mathbf{z}_i)}_{\text{topology-varying residual geometry}}.}$$

This separation allows human anatomy to act not merely as a coordinate reparameterization, but as an explicit inductive bias for learning a common functional 3D space across footwear whose geometry and topology may otherwise be radically different.